

Matthew Stephen Woodstock, Ph.D.

Computational Marine Biologist | Quantitative Ecosystem Modeler

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Education

Ph.D, Biological Sciences | Florida International University, Miami, FL (2022)

- Dissertation: *Ecological Modeling of the Oceanic Zone: A Gulf of Mexico Case Study*
- Advisor: Dr. Yuying Zhang

M.Sc., Marine Science | Nova Southeastern University, Dania Beach, FL (2018)

- Thesis: *Trophic Ecology and Parasitism of a Mesopelagic Fish Assemblage in the Gulf of Mexico*
- Advisor: Dr. Tracey Sutton

B.Sc., Ecology, Evolution, and Behavioral Biology | Beloit College, Beloit, WI (2015)

Professional Appointments

Assistant Scientist | University of Miami Cooperative Institute for Marine and Atmospheric Studies (2024-Present) | NOAA Southeast Fisheries Science Center

- *Gulf of Mexico Ecosystem Assessment*: Leading projects to evaluate climate stressors on economically important stocks, exploring ecosystem-based drivers of King Mackerel (*Scomberomorus cavalla*) dynamics to inform upcoming stock assessments, and improving remote computing capacity for ecosystem model calibration.
- *Lead Ecosystem Modeler, Florida Regional Ecosystem Stressors Collaborative Assessment (FRESCA)*: Developing a high-resolution Ecospace model for the southwest Florida region (including FL Keys and Florida Bay). Integrating downscaled regional climate projections (CMIP6) to predict habitat persistence and species distribution shifts under climate change scenarios. Managing an ecosystem modeling budget of ~\$240K throughout the project.
- *Principal Investigator, NCEAS Working Group*: Leading a multi-institutional team at the National Center for Ecological Analysis and Synthesis (NCEAS) to assess the ecological impacts of marine artificial structure implementation and decommissioning in the Gulf of Mexico. Coordinating three subgroups (systematic mapping, time series analysis, ecosystem modeling) and mentoring three NCEAS postdoctoral scholars (including two former FIU graduate students: Jonathan Rodemann and Nicholas Tucker).
- *Co-Developer, Valkyrie*: Collaborating with NOAA's Office of Science and Technology to develop a cloud-based agent-based model (ABM) for fisheries assessment. Integrating disparate databases (oceanographic models, trait databases) to predict fish bioenergetics, movement, and survival.

Postdoctoral Scholar | Woods Hole Oceanographic Institution (WHOI) (2023–2024)

- Developed *SwimmingIndividuals*, an open-source, high-performance agent-based marine ecosystem model written in the Julia programming language, that predicts population dynamics, distributions, and life histories from the integration of bioenergetic theory, movement ecology, behavioral dynamics, trophic ecology, and fisheries.
- Created pathways to integrate large (terabytes) oceanographic and biogeochemical model products into complex ecosystem models, and then additional pathways to output and analyze large datasets

- Represented the WHOI postdoc population by serving on the hiring committee for a successful Vice President of Academic Programs search
- Advisor: Dr. Gregory Britten

Postdoctoral Fellow | Morgan State University (2022–2023)

- Led the development of two ecosystem models (Ecopath with Ecosim) models in the Chesapeake Bay to predict the ecological and economic impacts of oyster restoration and Eelgrass (*Zostera marina*) die-off.
- Interviewed commercial fishers about their business expenditures to inform an economic impact assessment model, IMPLAN, to predict the direct, indirect, and induced economic impacts of restoration success outcomes in the Chesapeake Bay
- Mentored two graduate students through successful dissertation defenses, while also serving on their respective committees
- Advised by Dr. Thomas Ihde

Publications

Bibliometrics (current as of 2/6/26): Citations: 200 | h-index: 7 | i10-index: 7

Legend: **Bold** = Self

Peer-Reviewed Articles (In Review)

1. Tucker, N.A.C., T.N. Riley, L.M. Bradley, W. Klajbor, A.M. Knights, K. Mille, M. Steen, V. Stelzenmuller, A.B. Paxton, **M.S. Woodstock**. (In Review). What evidence exists on the ecological effects of marine artificial structures on fishes? A systematic map protocol. *PROCEED*.
2. **Woodstock, M.S.**, J.S. Bigman. (In Review). The scaling discrepancy between population consumption (Q/B) and individual daily ration. *Ecological Modelling*.
3. **Woodstock, M.S.**, J.P. Mattern, Z. Wu, G.L. Britten. (In Review). *SwimmingIndividuals: A high-performance agent-based model for marine ecosystems. PLOS Computational Biology*.
4. Iglesias, I., L. Kaikkonen, M. Oostdijk, L.G. Elsler, N. Yang, **M.S. Woodstock**, P. Annasawmy, K.M. Gjerde, R. Pallezo, P. Auster, E. Garcia-Seoane, E. Escobar-Briones, J.M.A. van der Grient, M. Gianni, R.J. Milligan, T. Sutton, B.M.J. Vastenhoud, I.A. Martins, T. Amaro, A.F. Johnson, A. Colaco, C.A. Choy, A.B. Leitner, A. Brandt, J.C. Drazen, L. Victorero. (In Review). Critical science and policy gaps limit our ability to evaluate anthropogenic impacts on global mesopelagic fishes. *Marine Policy*
5. Grassian, B., R. Domokos, P. Annasawmy, K. Boswell, T. Cariou, P. Escobar-Flores, S. Halfter, I. Iglesias, M. Jech, M. Laget, S. Hernandez-Leon, R. Milligan, T. Sutton, K. Witting, **M. Woodstock**. (In Review). The grand challenge of mesopelagic biomass estimation: Current strategies and recommendations. *Progress in Oceanography*

Peer-Reviewed Articles (Published)

6. **Woodstock, M.S.**, J.J. Kiszka, P.G.H. Evans, J.J. Waggitt, Y. Zhang. (2025). Marine mammal and seabird population changes have limited impacts on fisheries catches in the North Sea. *Canadian Journal of Fisheries and Aquatic Sciences*. 82: 1–14. [10.1139/cjfas-2025-0056](https://doi.org/10.1139/cjfas-2025-0056)
7. **Woodstock, M.S.**, H.E. Harris. (2025). Ecosystem models can predict the consequences of conservation management decisions. *Ecology Letters*. [10.1111/ele.70116](https://doi.org/10.1111/ele.70116)
8. Ferone, G., **M.S. Woodstock**, A. Hearn. (2025). Random Swims: An evaluation of acoustic telemetry thresholds for reef-shark behavior and residency. *Animal Biotelemetry*. [10.1186/s40317-025-00409-1](https://doi.org/10.1186/s40317-025-00409-1)

9. **Woodstock, M.S.**, A.T. Bevens*, M. Sulyman*, J. Maples, S. Knoche, T.F. Ihde. (2024). The economic impacts of living habitat change in the Virginia Middle Peninsula, Chesapeake Bay. *Ecological Modelling*. 498:110914. [10.1016/j.ecolmodel.2024.110914](https://doi.org/10.1016/j.ecolmodel.2024.110914)
10. Martin, A., A.B. Dominguez, C. Baker, C. Baumas, K. Bisson, E. Cavan, M. Freilich, E. Galbraith, M. Gali, S. Henson, K. Kvale, C. Lemmen, J. Luo, H. McMonagle, F. de Melo Virissimo, K. Ove Moller, C. Richon, I. Suresh, J. Wilson, **M.S. Woodstock**, A. Yool. (2024). When to add a new process to a model – and when not: a marine biogeochemical perspective. *Ecological Modelling*. 498:110870. [10.1016/j.ecolmodel.2024.110870](https://doi.org/10.1016/j.ecolmodel.2024.110870)
11. **Woodstock, M.S.**, J.J. Kiszka, M.R. Ramírez-León, T.T. Sutton, K. Fennel, B. Wang, Y. Zhang. (2023). Cetacean-mediated nitrogen transport in the oceanic Gulf of Mexico. *Limnology and Oceanography*. 68:2445–2460. [10.1002/lno.12433](https://doi.org/10.1002/lno.12433) (Featured in *L&O Bulletin*)
12. Richards, T., T.T. Sutton, **M.S. Woodstock**, H. Judkins, D. Wells. (2023). Body size, depth of occurrence, and local oceanography shape trophic structure in a diverse deep-pelagic micronekton assemblage. *Progress in Oceanography*. 213:102998. [10.1016/j.pocean.2023.102998](https://doi.org/10.1016/j.pocean.2023.102998)
13. **Woodstock, M.S.**, Y. Zhang. (2022). Towards ecosystem modeling in the deep sea: A review of past efforts and primer for the future. *Deep-Sea Research Part I*. 188:103851. [10.1016/j.dsr.2022.103851](https://doi.org/10.1016/j.dsr.2022.103851)
14. **Woodstock, M.S.**, T.T. Sutton, Y. Zhang. (2022). A trait-based carbon export model for mesopelagic fishes in the Gulf of Mexico with consideration of asynchronous vertical migration, flux boundaries, and feeding guilds. *Limnology and Oceanography*. 67:1443–1455. [10.1002/lno.12093](https://doi.org/10.1002/lno.12093)
15. Kiszka, J.J., **M.S. Woodstock**, M. Heithaus. (2022). Functional roles and ecological importance of small cetaceans in aquatic ecosystems. *Frontiers in Marine Science*. [10.3389/fmars.2022.803173](https://doi.org/10.3389/fmars.2022.803173)
16. **Woodstock, M.S.**, T.T. Sutton, T. Frank, Y. Zhang. (2021). An early warning sign: trophic structure changes in the oceanic Gulf of Mexico from 2011–2018. *Ecological Modelling*. 445:109509. [10.1016/j.ecolmodel.2021.109509](https://doi.org/10.1016/j.ecolmodel.2021.109509)
17. **Woodstock, M.S.**, C.A. Blamar, T.T. Sutton. (2020). Diet and parasites of a mesopelagic fish assemblage in the Gulf of Mexico. *Marine Biology*. 167:184. [10.1007/s00227-020-03796-6](https://doi.org/10.1007/s00227-020-03796-6)
18. Beck, H.N., A. Cohen, T. McKenzie, R. Weisend, K.W. Wikins, **M.S. Woodstock**. (2019). Broadening Horizons: Graduate Students Participating in International Collaborations. *Limnology and Oceanography Bulletin*. 28: 85–89. [10.1002/lob.10339](https://doi.org/10.1002/lob.10339)
19. **Woodstock, M.S.**, C. Golightly, D. Fenolio, J.A. Moore. (2019). *Larsonia pterophylla* parasitic on two Anguilliformes in the Gulf of Mexico. *Gulf and Caribbean Research*. 30:SC7–10. [10.18785/gcr.3001.05](https://doi.org/10.18785/gcr.3001.05)

Published Technical Reports

20. **Woodstock, M.S.**, S.R. Sagarese (2026). SEDAR 99-WP-05: Evaluating Environmental and Trophic Drivers of Gulf of America King Mackerel Using a Spatially Explicit Ecosystem Model. *Working Paper in the SEDAR99 King Mackerel stock assessment*
21. Harris, H.E., Sagarese S.R., **Woodstock M.S.**, and D.D. Chagaris. 2026. A U.S. Gulf-wide Ecopath with Ecosim and Ecospace model for the continental shelf incorporating spatial-temporal environmental drivers. NOAA Technical Memorandum NMFS-SEFSC-797. <https://doi.org/10.25923/zagd-vq81>

Manuscripts in Preparation (First author manuscripts only; manuscript available on request)

22. **Woodstock, M.S.**, H.E. Harris, D.D. Chagaris, S.R. Sagarese (In Prep). Accounting for Environmental Non-stationarity in Gulf of America Reef Fish Dynamics using a Spatially Explicit Ecosystem Model. *Intended for Fish and Fisheries*

23. **Woodstock, M.S.,** H.E. Harris, D.D. Chagaris, S.R. Sagarese (In Prep). Disentangling Environmental and Fishery Drivers of Penaeid Shrimp Dynamics in the U.S. Gulf of America. *Intended for Fisheries Oceanography*
24. **Woodstock, M.S.,** S.R. Sagarese (In Prep). Environmental and Trophic Drivers of a Coastal Migratory Pelagic Fish Population in the Gulf of America. *Intended for Fish and Fisheries*

Software and Data Products

SwimmingIndividuals: High-performance agent-based model (ABM) for marine ecosystems. Simulates metabolic and behavioral trajectories of millions of individuals using parallel, hybrid GPU-CPU computing.

- Development version hosted on Github:
<https://github.com/mattwoodstock/SwimmingIndividuals>
- Hosted on the NOAA Fisheries Integrated Toolbox: <https://nmfs-ost.github.io/noaa-fit/SI>

EcospaceCal: An R-based framework that automates the calibration of Ecopath with Ecosim (EwE) and Ecospace models using powerful modern optimization techniques.

- Development version hosted on Github: <https://github.com/mattwoodstock/EcospaceCal>

EwECaps: An R package that takes Ecopath with Ecosim (EwE) and Ecospace output and calculates ecosystem caps (a measure of biomass and production) to calculate maximum sustainable yields.

- Development version hosted on Github: <https://github.com/mattwoodstock/EwECaps>

Funded Proposals as PI (Cumulative: \$2.8M)

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| 2025 | Deepwater Horizon Open Ocean Trustee Implementation Group Deep-Sea Benefits - Outcomes of Mesophotic and Deep Benthic Community Restoration Monitoring and Adaptive Management Activity. <i>Funder:</i> Deepwater Horizon Trustee Council <i>Amount:</i> \$2.5M. <i>Role:</i> Principal Investigator. <i>Project Duration:</i> 2026–2028 |
| 2025 | Disentangling ecological impacts of marine artificial structure installation, decommissioning, and management scenarios on Gulf of America reef fish communities. <i>Funder:</i> NCEAS Gulf Ecosystem Initiative. <i>Amount:</i> \$125K. <i>Role:</i> Principal Investigator. <i>Project Duration:</i> 2026–2027 |
| 2023 | Development of an agent-based marine ecosystem model to explore spatiotemporal variability in food web dynamics in the Ocean Twilight Zone. <i>Funder:</i> Woods Hole Oceanographic Institution. <i>Amount:</i> \$125K. <i>Role:</i> Principal Investigator/Postdoctoral Scholar. <i>Project Duration:</i> 2023–2024. |

Recent Proposals Not Awarded

- Identifying and Validating the Effects of Shoreline Modification on Marine Ecosystem Structure and Function. *Funder:* NSF – HBCU Excellence in Research. *Full Project Amount:* \$614,896. *Transferable Project Amount:* \$59,976. *Role:* Co-Investigator. *Project Duration:* 2026–2029.
- Abundance by Design: Strategic habitat restoration through spatial fisheries modeling in the Florida Keys. *Funder:* EPA South Florida Program. *Full Project Amount:* \$499,625. *UM Project Amount:* \$210,985. *Role:* Co-Investigator. *Project Duration:* 2026-2030.

Teaching Experience

Guest Lecturer | Nova Southeastern University

- Delivered a lecture on the linkage between parasite communities and trophic ecology

Guest Lecturer | University of South Carolina

- Provided a lecture on my experience conducting research aboard an oceanographic research vessel

Teaching Assistant | Florida International University

- **General Biology II Lab (Ecology & Evolution):** (2022) Implemented a redesigned curriculum to integrate R-based data analysis with field collection. Led 50+ students in collecting biodiversity data and processing it using computational tools. Mentored 8 student-led independent research projects.
- **Human Biology Lab:** (2020) Managed online delivery of 12 laboratory modules for ~200 students.
- **General Biology I Lab:** (2018) Taught cellular processes and genetics; developed assessments.

Graduate Teaching Assistant | Nova Southeastern University

- **Graduate Ichthyology:** (2017) Assisted for 12 interactive lab exercises focused on the diversity and systematics of fishes. Curated the NSU fish collection, which included taking record of available specimens (~300 species) and replacing chemicals when necessary.

STEM Educator/Curriculum Developer | Museum of Discovery and Science (2017–2018)

- Developed and taught 50+ interactive STEM modules for K-12 students, focusing on data literacy and marine science.

Mentoring Experience

Graduate Committees

- **Nicole Luchau** (M.S., University of Florida) – Current student
- **Amanda Bevans** (Ph.D., Morgan State University) – Now Postdoc at Morgan State University
- **Muhammad Sulyman** (Ph.D., Morgan State University) – Now Lecturer at Federal University, Nigeria

Other Experience

- Hollings Scholar Co-Mentor (2026; upcoming) – Marisol Helms (UC Santa Barbara)
- FIU Graduate Student Mentor (2021)
- Beloit College Alumni Mentoring Network (2020)
- Nova Southeastern University Graduate Student Mentor (2016)

Presentations

Legend: **Bold** = Self | * = Mentored Graduate Students | † = Mentored Undergraduate Students & Technicians

Invited Seminars

1. *Chesapeake Research Consortium* (2025): "Economic impacts of oyster restoration."
2. *Oregon State University* (2024): "Insights into deep-pelagic energetics with mechanistic models."
3. *WHOI Ocean Twilight Zone Symposium* (2023): "Food-web modeling in the mesopelagic zone."
4. *Chesapeake Bay Fisheries Goal Implementation Team Bi-annual Meeting* (2023): "The economic impacts of living habitat changes in the Virginia Middle Peninsula."
5. *Morgan State University Summer Internship Program* (2023): "Ecological modeling: discovering the world through computers"

Oral Conference Presentations

6. Helms, M.[†], R. Klasky, **M.S. Woodstock**, J. Bigman, J. Quinlan, H. Townsend. (upcoming: Feb. 2027). Consideration of agent-based fleet dynamics in an agent-based ecological model, *Valkyrie*. ASLO Aquatic Sciences Meeting. San Juan, Puerto Rico.
7. Helms, M.[†], R. Klasky, **M.S. Woodstock**, J. Bigman, J. Quinlan, H. Townsend. (upcoming: Nov. 2026). Visualizing model output for fisheries management decision-making. Mid-Atlantic Regional American Fisheries Society Annual Meeting. Cape May, NJ.
8. Helms, M.[†], R. Klasky, **M.S. Woodstock**, J. Bigman, J. Quinlan, H. Townsend. (2026). Visualizing model output for fisheries management decision-making. NOAA Hollings Scholar symposium. Silver Spring, MD.
9. Karp, M.A., B. Li, A. Havron, J. Bigman, **M.S. Woodstock**, D. Chagaris, K. Shotwell. (2026). Evaluating the relative performance of using random effects versus environmental covariates in single-species stock assessment models. ICES Annual Science Conference. Brest, France.
10. Havron, A., M.A. Karp, B. Li, J. Bigman, **M.S. Woodstock**, D. Chagaris, K. Shotwell. (2026). Evaluating the relative predictive performance of random effects versus environmental covariates in fisheries production models. International Statistical Ecology Conference. Mérida, Mexico.
11. Bevens, A.T.* , M. Sulyman* , **M.S. Woodstock**, J. Maples, S. Knoche, T.F. Ihde. (2025). Modeling ecological and economic impacts of living habitat change in the York River, Chesapeake Bay. Coastal and Estuarine Research Foundation Biennial Conference. Richmond, VA.
12. Sutton, T., K. Boswell, H. Bracken-Grissom, A. Cook, T. Frank, M. Johnston, H. Judkins, R. Milligan, J. Moore, P. Peres, J. Quinlan, I. Romero, M. Vecchione, R. Bos, A. Millett, T. Richards, **M.S. Woodstock**, Y. Zhang, I. Zink. (2025). Deep-Pelagic Nekton Research in the Gulf of Mexico: a 14-year synopsis and future directions. Deep-Sea Biology Society Meeting. Hong Kong SAR, China.
13. Bevens, A.B.* , M. Sulyman* , **M.S. Woodstock**, S. Knoche, T. Ihde. (2024). The ecological and economic effects of habitat change in the Chesapeake Bay. American Fisheries Society National Meeting. Honolulu, HI.
14. **M.S. Woodstock**, J.P. Mattern, Z. Wu, G.L. Britten. (2024). Climate change impacts the individual energy budgets of oceanic fishes. American Fisheries Society National Meeting, Honolulu, HI.
15. Bevens, A.T.* , M. Sulyman* , **M.S. Woodstock**, S. Knoche, T. Ihde. (2024). Modeling the effects of the habitat change in the York and Piankatank Rivers. Chesapeake Community Research Symposium. Annapolis, MD.
16. Bevens, A.T.* , M. Sulyman* , **M.S. Woodstock**, S. Knoche, T. Ihde. (2024). Understanding foundational habitat changes in the Chesapeake Bay with a York River ecosystem model. Atlantic Estuarine Research Society. Gloucester Point, VA.
17. Hoyt, E.[†], **M.S. Woodstock**, P. Piavis, D. Sanderson-Kilchenstein, T.F. Ihde. (2023). The effects of spatial bias on the management of fished populations. Chesapeake Watershed Forum. Shepardsdown, WV.
18. Sulyman, M.* , A.T. Bevens* , **M.S. Woodstock**, S. Knoche, T. Ihde. (2023). The ecological impacts of oyster reef restoration in Virginia's Middle Peninsula. Chesapeake Oyster Science Symposium, Cambridge, MD.

19. **Woodstock, M.S.**, A.T. Bevans*, M. Sulyman*, Y. Zhang, J.J. Kiszka, T. Ihde. (2023). Predicting causality in ecosystem models using empirical dynamic modeling. American Fisheries Society National Meeting. Grand Rapids, MI.
20. Bevans, A.T.*, M. Sulyman*, **M.S. Woodstock**, S. Knoche, T. Ihde. (2023). Modeling habitat change effects in the York River ecosystem, Chesapeake Bay. American Fisheries Society National Meeting. Grand Rapids, MI.
21. **Woodstock, M.S.**, A.T. Bevans*, M. Sulyman*, S. Knoche, T. Ihde. (2023). The ecosystem-scale impacts of living habitat change in Virginia's Middle Peninsula. American Fisheries Society National Meeting. Grand Rapids, MI.
22. Bevans, A.T.*, M. Sulyman*, **M.S. Woodstock**, S. Knoche, T. Ihde. (2023). Improving our understanding of habitat changes in the Chesapeake Bay with a York River ecosystem model. Morgan State University Spring Into Research Graduate Symposium. Baltimore, MD.
23. Bevans, A.T.*, M. Sulyman*, **M.S. Woodstock**, S. Knoche, T. Ihde. (2022). Estimating the Commercial Fishing Effects of Habitat Restoration in Virginia's Middle Peninsula. Chesapeake Oyster Alliance Meeting.
24. Sutton, T., Z. Milligan, K. Boswell, H. Bracken-Grissom, A. Cook, T. Frank, D. Hahn, M. Johnston, H. Judkins, J. Moore, J. Moore, J. Quinlan, P. Peres, I. Romero, M. Vecchione, M. Shivji, A. Bernard, M. D'Elia, **M. Woodstock**, Y. Zhang, K. Benson, M. Karnauskas, F. Parker. (2022). The state of the pelagic Gulf of Mexico: the continuing mission of the DEEPEND research consortium. The Gulf of Mexico Conference.
25. **Woodstock, M.S.**, T.T. Sutton, Y. Zhang. (2022). A trait-based carbon export model for mesopelagic fishes in the Gulf of Mexico with consideration of asynchronous vertical migration, flux boundaries, and feeding guilds. ASLO Ocean Sciences Meeting. Online Presentation.
26. **Woodstock, M.S.** (2021). SciComm beyond LOREX: How an international research program inspired future science communication efforts. ASLO Aquatic Sciences Meeting. Online Presentation.
27. **Woodstock, M.S.**, T.T. Sutton, T. Frank, Y. Zhang. (2021). Assessing trophic structure dynamics in ecosystem models using the offshore Gulf of Mexico as an example. ASLO Aquatic Sciences Meeting. Online Presentation.
28. **Woodstock, M.S.**, J.J. Kiszka, P.G.H. Evans, J.J. Waggitt, Y. Zhang. (2021). Debunking Misconceptions: Marine mammals and seabirds have limited impacts on fisheries catches in the North Sea. Florida International University Biosymposium. Online Presentation.
29. **Woodstock, M.S.**, T.T. Sutton, T. Frank, Y. Zhang. (2020). Assessing trophic structure dynamics in ecosystem models using the offshore Gulf of Mexico as an example. American Fisheries Society Annual Meeting. Online Presentation.
30. **Woodstock, M.S.**, T.T. Sutton, T. Frank, Y. Zhang. (2020). An early warning sign: trophic structure changes in the oceanic Gulf of Mexico from 2011–2018. Deep Sea Biology Society Meeting. Online Presentation.
31. Chua, E., E. Knotts, K. Wilkins, **M.S. Woodstock**, A.R. Marín. (2020). Limnology and Oceanography Research Exchange (LOREX). European Geosciences Union Meeting. Online Presentation.
32. **Woodstock, M.S.** (2020). Take a hike: creating a positive work-life balance through excursions. Ocean Sciences Meeting. San Diego, CA.

33. **Woodstock, M.S.**, B. Wang, K. Fennel, T.T. Sutton, Y. Zhang. (2020). Ecological importance of mesopelagic fishes in the oceanic Gulf of Mexico. Florida International University Biosymposium. North Miami, FL.
34. Quiquempois, V., **M.S. Woodstock**, Y. Zhang, M. Heithaus, J. Kiszka. (2019). The top-down effects of cetaceans in the Gulf of Mexico: Who are the key players? World Marine Mammal Conference. Barcelona, Spain.
35. **Woodstock, M.S.** (2019). Food web model of the oceanic Gulf of Mexico. Florida International University Biosymposium. North Miami, FL.
36. **Woodstock, M.S.**, C.A. Blonar, T.T. Sutton. (2018). Trophic ecology and parasitism of a mesopelagic fish assemblage. Nova Southeastern University Biosymposium. Dania Beach, FL.
37. **Woodstock, M.S.**, C.A. Blonar, T.T. Sutton. (2017). An examination of the parasites and trophic ecology of mesopelagic fishes. American Fisheries Society Annual Meeting. Tampa, FL.
38. **Woodstock, M.S.**, C.A. Blonar, T.T. Sutton (2017). An examination of the parasites and trophic ecology of mesopelagic fishes. American Society of Ichthyologists and Herpetologists Annual Meeting. Austin, TX.

Poster Conference Presentations

39. Iglesias, I., L. Kaikkonen, N. Yang, **M.S. Woodstock**, M. Oostdijk, L. Elsler, A.P. Annasawmy, L. Victorero. (2025). Science-policy gap analysis for mesopelagic fishes, fisheries, and ecosystems. Deep-Sea Biology Society Meeting. Hong Kong SAR, China.
40. **Woodstock, M.S.**, G.L. Britten. (2024). An individual-based model for an energetic cost-benefit analysis of diel vertical migration. CLIVAR Pathways Connecting Climate Changes to the Deep Ocean Workshop. Lewes, DE.
41. **Woodstock, M.S.**, J.J. Kiszka, M. Rafael Ramirez-Leon, T.T. Sutton, K. Fennel, B. Wang, Y. Zhang. (2024). Cetacean-mediated vertical nitrogen transport in the oceanic Gulf of Mexico. Gulf of Mexico Conference. Tampa, FL.
42. **Woodstock, M.S.**, G.L. Britten. (2023). A novel, individual-based ecosystem model for fine-scale food-web processes. WHOI Ocean Twilight Zone Symposium. Woods Hole, MA.
43. Sulyman, M.*, A.T. Bevans*, **M.S. Woodstock**, S. Knoche, T.F. Ihde. Modeling changes to foundational habitats in the Piankatank River, Chesapeake Bay. Atlantic Estuarine Research Society. Gloucester Point, VA.
44. Bevans, A.T.*, M. Sulyman*, **M.S. Woodstock**, S. Knoche, T. Ihde. (2023). Modeling the effects of habitat changes in the York River ecosystem, Chesapeake Bay. Shepardstown, WV.
45. Hoyt, E.E.[†], **M.S. Woodstock**, P. Piavis, D. Sanderson-Kilchenstein, T. Ihde. (2023). An exploration of natural mortality and maximum age for Chesapeake Bay fishes. American Fisheries Society National Meeting. Grand Rapids, MI.
46. Bevans, A.T.*, M. Sulyman*, **M.S. Woodstock**, S. Knoche, T. Ihde. (2023). Improving Our Understanding of Habitat Changes in the Chesapeake Bay with a York River Ecosystem Model. York River Science Symposium. Gloucester, VA.
47. Sulyman, M.*, A.T. Bevans*, **M.S. Woodstock**, S. Knoche, T. Ihde. (2023). Virginia's Middle Peninsula Habitat Focus Area: The Economic Impacts of Resource Restoration and Habitat Change. Morgan State University Spring Into Research Graduate Symposium. Baltimore, MD.
48. Hoyt, E.E.[†], **M.S. Woodstock**, T.F. Ihde. (2023). An exploration of age and growth estimates for fishes in the Chesapeake Bay. American Fisheries Society Tidewater Chapter Meeting. Solomon's Island, MD.

49. Bevens, A.T.*, M. Sulyman*, **M.S. Woodstock**, S. Knoche, T.F. Ihde. Improving our understanding of habitat changes in the Chesapeake Bay with a York River ecosystem model. American Fisheries Society Tidewater Chapter Meeting. Solomon's Island, MD.
50. Hoyt, E.E.[†], **M.S. Woodstock**, T.F. Ihde. (2022). A case for tributary-specific age/growth estimates for fishes in the Chesapeake Bay. Southern Maryland Marine Science Symposium. St. Mary's, MD.
51. Bevens, A.T.*, Sulyman, M.* , **M.S. Woodstock**, S. Knoche, T.F. Ihde. (2022). Virginia's Middle Peninsula Habitat Focus Area: the Economic Impacts of Resource Restoration and Habitat Change. Southern Maryland Marine Science Symposium. St. Mary's, MD.
52. Hoyt, E.E.[†], **M.S. Woodstock**, T.F. Ihde. (2022). A case for tributary-specific age/growth estimates for fishes in the Chesapeake Bay. Chesapeake Watershed Forum. Shepardstown, WV.
53. Sulyman, M.*, A.T. Bevens*, **M.S. Woodstock**, S. Knoche, T.F. Ihde. (2022). Virginia's Middle Peninsula Habitat Focus Area: the Economic Impacts of Resource Restoration and Habitat Change. Chesapeake Watershed Forum. Shepardstown, WV.
54. **Woodstock, M.S.**, J.J. Kizka, P.G.H. Evans, J.J. Waggitt, Y. Zhang. (2022). Debunking misconceptions: rising marine mammal abundances have little impact on fisheries in the southern North Sea. 24th Biennial Conference on the Biology of Marine Mammals. Palm Beach, FL.
55. Sutton, T.T., A.B. Cook, K.M. Boswell, H.D. Bracken-Grissom, R. Eytan, D. Fenolio, T. Frank, D. Hahn, M.W. Johnston, H. Judkins, R.J. Milligan, J. Moore, J. Quinlan, T. Richards, I.C. Romero, M. Shivji, A. Bernard, M. Vecchione, **M.S. Woodstock**, Y. Zhang. (2022). Sustained observation of the deep-pelagic Gulf of Mexico: the DEEPEND|RESTORE program. ASLO Ocean Sciences Meeting. Online Presentation.
56. **Woodstock, M.S.**, T.T. Sutton, T. Frank, Y. Zhang. (2021). Ecosystem modeling in the oceanic zone: A Gulf of Mexico case study. 16th Deep Sea Biology Symposium. Brest, France.
57. **Woodstock, M.S.**, B. Wang, K. Fennel, T.T. Sutton, Y. Zhang (2020). A comparison of two ecosystem models of the oceanic Gulf of Mexico. Ocean Sciences Meeting. San Diego, CA.
58. **Woodstock, M.S.**, C.A. Blonar, T.T. Sutton. (2018). On parasitism in mesopelagic fishes as a function of trophic ecology and vertical distribution. Deep Sea Biological Society Meeting. Monterey Bay, CA.
59. **Woodstock, M.S.**, T.T. Sutton, C.A. Blonar. (2018). Trophic ecology and parasitism of a deep-pelagic fish assemblage. American Society of Parasitologists Annual Meeting. Cancun, Mexico.
60. **Woodstock, M.S.**, C.A. Blonar, T.T. Sutton. (2017). Variations in the parasite fauna and gut contents of vertically migrating and non-migrating mesopelagic fishes of the northern Gulf of Mexico. Gulf of Mexico Oil Spill and Ecosystem Science Conference. New Orleans, LA.

Awards and Other Funding (Cumulative: \$164,850)

2026	Southwest Fisheries Science Center Team Member of the Year
2024	OCB-CLIVAR Pathways Conference Early Career Travel Award (\$500)
2023	Ocean Twilight Zone-Jetzon Symposium Early Career Travel Award (\$500)
2022	FIU Provost Employer Supported Tuition Fellowship (\$3,000)
2021	FIU Coastlines and Oceans Division Travel Award (\$500)
2021	FIU Biosymposium 2 nd Best Oral Presentation (\$50)
2020	AFS Florida Chapter Student Subunit Travel Award (\$100)

2019 FIU College of Arts and Sciences Travel Award (\$300)
 2019 FIU Professional Development Grant (\$300)
 2019 Limnology and Oceanography Research Exchange (LOREX) Scholar (\$4,000)
 2018 NSU Halmos College of Natural Sciences and Oceanography Student of the Year
 2017 NSU Professional Development Grant (\$600)
 2016 NSU Oceanographic Center Fishing Tournament Scholarship (\$30,000)
 2015 Academic All-Midwest Conference Baseball Team
 2015 Beloit College Dean's List
 2014 Beloit College Dean's List

Professional Service

Editorial Service

- **Associate Editor**, *Food Webs* (2024–Present)

Grant & Manuscript Review

- **Fellowship Reviewer:** NMFS-Sea Grant Population and Ecosystem Dynamics Fellowship (2025); NSF Biological Oceanography Ad hoc Reviewer (2025); American Fisheries Society (AFS) Hutton Scholarship (2021–2025)
- **Journal Reviewer (number of reviews):** *BMC Marine Science*, *Deep-Sea Research*, *Ecological Modelling* (7), *Estuaries and Coasts*, *Food Webs* (2), *Functional Ecology*, *ICES Journal of Marine Science* (2), *Journal of Fish Biology*, *Journal of Plankton Research*, *Journal of Zoology*, *Marine Biology* (2), *Ocean and Coastal Management*, *Ocean Sustainability*, *One Earth*

Leadership

- **Session/Meeting Organizer:** Deep-Sea Biology Society Meeting (2025; could not attend); American Fisheries Society National Meeting (2023); FIU Biosymposium (2021); FIU Marine Science Seminar Series (2019)
- **Working Group Member:** JETZON Biological Carbon Pump Working Group (2023–Present); JETZON Early Career Researcher Working Group (2023–Present); Deep-Ocean Stewardship Initiative (DOSI) Open-Access Task Force (2021); DOSI Fisheries Working Group (2020–Present)
- **Professional Trainings:** Ocean Exploration Cooperative Institute Facilitation Training (2024)
- **Search Committee:** WHOI VP/Dean of Academic Programs Search Committee (2024)
- **Other Leadership Examples:** Subject matter expert on several NOAA-internal environmental impact statement documents (2025–Present); FIU Biology Graduate Student Committee Officer (2020); American Fisheries Society Florida Student Subunit Chapter Officer; Alpha Zeta Chapter of Sigma Chi (2012–2015); Beloit College Varsity Baseball (2012–2015)

Oceanographic Research Cruise Experience

R/V *Point Sur*, Gulf of Mexico (2021; 10 days at sea)

- Coordinated and twice-daily CTD operations resulting in the collection of localized physical, biogeochemical, and biological data, as well as the collection of water for petrochemical and eDNA analyses
- Led the deployment and data collection for broadband and wideband acoustic transmitter for the collection of bioacoustics data at coarse and individual scales
- Coordinated with the MOCNESS net operator and ship captain about acoustic-derived depths for last-minute adjustments to sampling regimes

- Mentored a graduate student in CTD operations

R/V *Weatherbird*, Florida Straits (2019; 3 days at sea)

- Coordinated Tucker trawl midwater net deployments
- Instructed other cruise participants on the identification of oceanic fishes, crustaceans, and cephalopods

R/V *Point Sur*, Gulf of Mexico (2017; 14 days at sea)

- Gathered biological data from MOCNESS trawl catches and assisted in database management
- Preserved specimens in appropriate fixation materials for subsequent stable isotope, metabarcoding, parasite, gut content, age/growth, and reproduction studies

Professional Societies (Past and Present)

Association for the Sciences of Limnology and Oceanography (ASLO), American Fisheries Society (AFS), American Institute for Fisheries and Research Biologists (AIFRB), American Society of Ichthyologists and Herpetologists (ASIH), Deep Sea Biological Society (DSBS), Society for Marine Mammalogy (SMM)